

Package ‘APCinteraction’

February 9, 2026

Type Package

Title All Possible Comparisons (APC) Nonparametric Tests for
Interaction in Balanced Two-Way ANOVA Models

Version 0.1.0

Description Provides novel nonparametric tests, ‘APCSSA’ and ‘APCSSM’, for interaction in two-way ANOVA designs with balanced replications using all possible comparisons. These statistics extend previous methods, allow greater flexibility, and demonstrate higher power in detecting interactions for non-normal data. The package includes optimized functions for computing these test statistics, generating interaction plots, and simulating their null distributions.

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Encoding UTF-8

URL <https://github.com/tranbaokhue/APCinteraction>

BugReports <https://github.com/tranbaokhue/APCinteraction/issues>

RoxygenNote 7.3.3

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0)

Imports magrittr, dplyr, parallel, pbapply, rlang

Suggests APCinteractionData, testthat (>= 3.0.0)

Config/testthat/edition 3

NeedsCompilation no

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APCinteraction

APCinteraction: All Possible Comparisons (APC) Nonparametric Tests for Interaction in Balanced Two-way ANOVA Models

Description

Provides nonparametric tests for interaction in two-way ANOVA designs with balanced replications using All Possible Comparisons (APC). The APCSSA and APCSSM statistics extend previous methods, allow greater flexibility, and demonstrate higher power in detecting interactions for non-normal data. The package includes optimized functions for computing these test statistics, generating interaction plots, and simulating their null distributions.

Main functions

[APCSSA](#) Compute the APCSSA interaction test statistic.

[sim_nullAPCSSA](#) Simulate the null distribution for the APCSSA statistic.

[APCSSM](#) Compute the APCSSM interaction test statistic.

[sim_nullAPCSSM](#) Simulate the null distribution for the APCSSM statistic.

[save_null](#) Save the simulated null distributions locally for future usage.

Author(s)

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See Also

Useful links:

- <https://github.com/tranbaokhue/APCinteraction>
- Report bugs at <https://github.com/tranbaokhue/APCinteraction/issues>

APCSSA

APCSSA

Description

This function tests for interaction in a two-way layout using all possible crossed comparisons (APC) based on aligned ranks, with alignment performed using the mean.

Usage

```
APCSSA(formula, data, numSim = 1e+05)
```

Arguments

formula	A formula specifying the model, with one response variable and two experimental factors.
data	A data frame in long format: the first column contains observed values of the response variable, the second column contains the levels of Factor A, and the third column contains the levels of Factor B.
numSim	An integer specifying the number of simulations used to estimate the null distribution. Defaults to 100,000 (1e+05). Even though this default might take longer to run for designs with factors of more than 5 levels, we recommend using at least 100,000 simulations to get reliable results.

Details

APCSSA performs a nonparametric test for interaction in a two-way layout with balanced replication. Two statistics, APCCRA and APCRCA, are computed by aligning the data with the averages and ranking within the rows or columns.

These statistics use all possible crossed comparisons to detect interaction effects. The final test statistic is the maximum of the two standardized statistics. A p -value is estimated by comparing this statistic to a pre-simulated null distribution specific to the design dimensions.

The companion package **APCinteractionData** provides 90 pre-computed APCSSA null distributions (100,000 simulations each) covering two-way designs with 2 to 6 factor levels and 1 to 10 replications. Install it with `devtools::install_github("tranbaokhue/APCinteractionData")` to enable fast p -value estimation for these designs.

If the design is not covered by the pre-computed distributions or if a higher precision is desired (`numSim > 100,000`), users must first generate the null distribution using `sim_nullAPCSSA` before using this function to evaluate the interaction effect.

For more information, see the referenced article.

Value

A data frame with the following columns:

- **Statistic:** The APCSSA test statistic (maximum of APCCRA and APCRCA) is provided.
- **p -value:** The estimated p -value is displayed.

A summary table is printed to the console, and an interaction plot is generated.

References

Tran, B. K., Wagaman, A. S., Nguyen, A., Jacobson, D., & Hartlaub, B. (2024). Nonparametric tests for interaction in two-way ANOVA with balanced replications. *arXiv preprint* arXiv:2410.04700. <https://arxiv.org/abs/2410.04700>

See Also

[sim_nullAPCSSA](#), [APCSSM](#)

Examples

```
# Set the seed for reproducibility
set.seed(206)

# Parameters - number of levels for factors A and B, and replications
nA <- 3; nB <- 3; nrep <- 3

# Generate levels and create the full design
A_vals <- 1:nA
B_vals <- 1:nB
design <- expand.grid(A = A_vals, B = B_vals)
design <- design[rep(seq_len(nrow(design)), each = nrep), ]

# Compute response as interaction (product of numeric factor levels)
mu <- A_vals[design$A] * B_vals[design$B]
value <- mu + rnorm(length(mu)) # Add normal noise

# Assemble the data
data <- data.frame(
  value = value,
  A = factor(design$A),
  B = factor(design$B)
)

# Run the APCSSA test
APCSSA(value ~ A + B, data = data)
```

APCSSM

APCSSM

Description

This function tests for interaction in a two-way layout using all possible crossed comparisons (APC) based on aligned ranks, with alignment performed using the median.

Usage

```
APCSSM(formula, data, numSim = 1e+05)
```

Arguments

formula	A formula specifying the model, with one response variable and two experimental factors.
data	A data frame in long format: the first column contains observed values of the response variable, the second column contains the levels of Factor A, and the third column contains the levels of Factor B.
numSim	An integer specifying the number of simulations used to estimate the null distribution. Defaults to 100,000 (1e+05). Even though this default might take longer to run for designs with factors of more than 5 levels, we recommend using at least 100,000 simulations to get reliable results.

Details

APCSSM performs a nonparametric test for interaction in a two-way layout with balanced replication. Two statistics, APCCRM and APCRCM, are computed by aligning the data with the medians and ranking within the rows or columns.

These statistics use all possible crossed comparisons to detect interaction effects. The final test statistic is the maximum of the two standardized statistics. A p -value is estimated by comparing this statistic to a pre-simulated null distribution specific to the design dimensions.

The companion package **APCinteractionData** provides 90 pre-computed APCSSM null distributions (100,000 simulations each) covering two-way designs with 2 to 6 factor levels and 1 to 10 replications. Install it with devtools: `install_github("tranbaokhue/APCinteractionData")` to enable fast p -value estimation for these designs.

If the design is not covered by the pre-computed distributions or if a higher precision is desired (`numSim > 100,000`), users must first generate the null distribution using `sim_nullAPCSSM` before using this function to evaluate the interaction effect.

For more information, see the referenced article.

Value

A data frame with the following columns:

- **Statistic:** The APCSSM test statistic (maximum of APCCRM and APCRCM) is provided.
- **p -value:** The estimated p -value is displayed.

A summary table is printed to the console, and an interaction plot is generated.

References

Tran, B. K., Wagaman, A. S., Nguyen, A., Jacobson, D., & Hartlaub, B. (2024). Nonparametric tests for interaction in two-way ANOVA with balanced replications. *arXiv preprint* arXiv:2410.04700. <https://arxiv.org/abs/2410.04700>

See Also

`sim_nullAPCSSM`, `APCSSA`

Examples

```
# Set seed for reproducibility
set.seed(36)

# Parameters - number of levels for factors A and B, replications, and interaction effect
nA <- 3; nB <- 4; nrep <- 2; c <- 1.25

# Generate levels and create the full design
A_vals <- seq(-2, 2, length.out = nA)
B_vals <- seq(-1.5, 1.5, length.out = nB)
design <- expand.grid(A = 1:nA, B = 1:nB)
design <- design[rep(seq_len(nrow(design)), each = nrep), ]

# Create the specific interaction matrix: alternate ±c in top rows
specInt <- matrix(0, nA, nB)
specInt[1:2, 1:nB] <- matrix(c(c, -c, -c, c), nrow = 2, byrow = TRUE)
```

```

# Compute response
mu <- A_vals[design$A] + B_vals[design$B] + specInt[cbind(design$A, design$B)]
value <- mu + rt(length(mu), df = 1)

# Assemble the data
data <- data.frame(
  value = value,
  A = factor(design$A, labels = round(A_vals, 2)),
  B = factor(design$B, labels = round(B_vals, 2))
)

# With Cauchy errors, we opt for APCSSM to check for interaction
APCSSM(value ~ A + B, data = data)

```

save_null

Save null distribution to working directory

Description

Saves a null distribution from the package's cache to an RData file in the working directory.

Usage

```
save_null(type, i, j, k, path = NULL)
```

Arguments

type	Character string: either "APCSSA" or "APCSSM."
i	The number of levels for factor A.
j	The number of levels for factor B.
k	The number of replications at each combination of Factor A and Factor B.
path	Optional custom file path. If NULL (default), saves to working directory.

Value

Invisibly returns the file path to where the data were saved.

See Also

[sim_nullAPCSSA](#), [sim_nullAPCSSM](#)

Examples

```

## Not run:
# After running simulation
sim_nullAPCSSA(2, 2, 2)

# Save to working directory
save_null("APCSSA", 2, 2, 2)

# Or save to custom location

```

```
save_null("APCSSA", 2, 2, 2, path = "~/my_nulls/")

## End(Not run)
```

sim_nullAPCSSA	<i>Simulate the null distribution for APCSSA</i>
----------------	--

Description

A function for simulating the null distribution of APCSSA.

Usage

```
sim_nullAPCSSA(i, j, k, numSim = 1e+05, parallel = TRUE, verbose = TRUE)
```

Arguments

<code>i</code>	The number of levels for factor A.
<code>j</code>	The number of levels for factor B.
<code>k</code>	The number of replications at each combination of Factor A and Factor B.
<code>numSim</code>	The number of simulations (data sets) to generate. Defaults to 100,000 (1e+05). Even though this default might take longer to run for designs with factors of more than 5 levels, we recommend using at least 100,000 simulations to get reliable results.
<code>parallel</code>	Boolean; if TRUE (the default), uses one fewer than the total number of cores for parallel computation.
<code>verbose</code>	Logical; if TRUE (the default), prints a startup notice and progress bar.

Details

While we have provided null distributions for various settings in this package, there are still two-way settings that we don't readily have the null distribution to evaluate the significance of the test statistics (see **References** for the complete list of settings available). For example, one might wish to estimate the p -value with a higher `numSim`, e.g., 250,000.

Accompanying this function is our customized [save_null](#) to help researchers save the null distribution generated for future reference and calculations of the statistics in similar experiment design.

Value

A data frame with the null mean and standard deviation for the two test statistics (APCCRA and APCRCA) that will get standardized into APCSSA and a numeric vector with length equal to the `numSim` of all APCSSA statistics on the null data sets.

Note

Benchmarks for 100,000 simulations (no bias toward any chip type):

- **3×3×3** (243 summands): ~ 5 hours on an Intel i5
- **2×5×5** (625 summands): < 5 hours on Apple M1 (7 cores, parallel)
- **6×6×5** (9375 summands): ~ 108 hours on Apple M1 (7 cores, parallel) Factor levels (i,j) have some impact, but total summands scale as k^4 (see **References** for more information), so replication depth (k) dominates run time.

Tip: leave `parallel = TRUE` to shorten run times dramatically.

Results are stored in the package's cache. The customized function `save_null` is highly recommended for saving the null distribution to your local storage for future reference and testing in the same design.

References

Tran, B. K., Wagaman, A. S., Nguyen, A., Jacobson, D., & Hartlaub, B. (2024). Nonparametric tests for interaction in two-way ANOVA with balanced replications. *arXiv preprint* arXiv:2410.04700. <https://arxiv.org/abs/2410.04700>

See Also

`save_null`, `APCSSA`, `sim_nullAPCSSM`

Examples

```
## Not run:
sim_nullAPCSSA(2, 2, 2, 5000) # This should take only a few seconds

## End(Not run)
```

sim_nullAPCSSM

Simulate the null distribution for APCSSM

Description

A function for simulating the null distribution of APCSSM

Usage

```
sim_nullAPCSSM(i, j, k, numSim = 1e+05, parallel = TRUE, verbose = TRUE)
```

Arguments

i	The number of levels for factor A.
j	The number of levels for factor B.
k	The number of replications at each combination of Factor A and Factor B.

numSim	The number of simulations (data sets) to generate. Defaults to 100,000 (1e+05). Even though this default might take longer to run for designs with factors of more than 5 levels, we recommend using at least 100,000 simulations to get reliable results.
parallel	Boolean; if TRUE (the default), uses one fewer than the total number of cores for parallel computation.
verbose	Logical; if TRUE (the default), prints a startup notice and progress bar.

Details

While we have provided null distributions for various settings in this package, there are still two-way settings that we don't readily have the null distribution to evaluate the significance of the test statistics (see **References** for the complete list of settings available). For example, one might wish to estimate the p -value with a higher numSim, e.g., 250,000.

#' Accompanying this function is our customized [save_null](#) to help researchers save the null distribution generated for future reference and calculations of the statistics in similar experiment design.

Value

A data frame with the null mean and standard deviation for the two test statistics (APCCRM and APCRCM) that will get standardized into APCSSM and a numeric vector with length equal to the numSim of all APCSSM statistics on the null data sets.

Note

Benchmarks for 100,000 simulations (no bias toward any chip type):

- **3×3×3** (243 summands): ~5 hours on an Intel i5
- **2×5×5** (625 summands): < 5 hours on Apple M1 (7 cores, parallel)
- **6×6×5** (9375 summands): ~108 hours on Apple M1 (7 cores, parallel) Factor levels (i,j) have some impact, but total summands scale as k^4 (see **References** for more information), so replication depth (k) dominates run time.

Tip: leave parallel = TRUE to shorten run times dramatically.

Results are stored in the package's cache. The customized function [save_null](#) is highly recommended for saving the null distribution to your local storage for future reference and testing in the same design.

References

Tran, B. K., Wagaman, A. S., Nguyen, A., Jacobson, D., & Hartlaub, B. (2024). Nonparametric tests for interaction in two-way ANOVA with balanced replications. *arXiv preprint* arXiv:2410.04700. <https://arxiv.org/abs/2410.04700>

See Also

[save_null](#), [APCSSM](#), [sim_nullAPCSSA](#)

Examples

```
## Not run:  
sim_nullAPCSSM(2, 2, 2, 5000) # This should take only a few seconds  
  
## End(Not run)
```

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